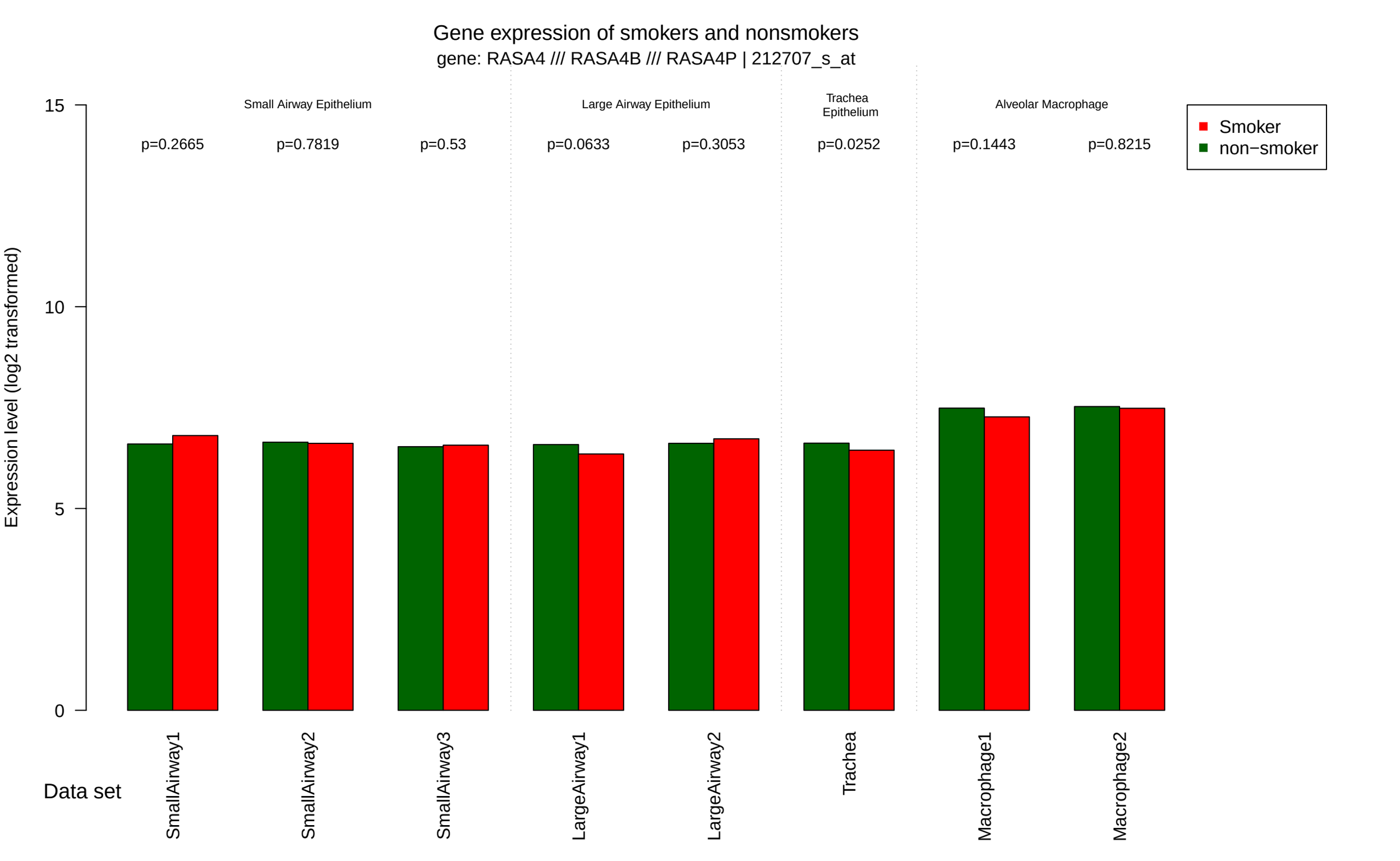
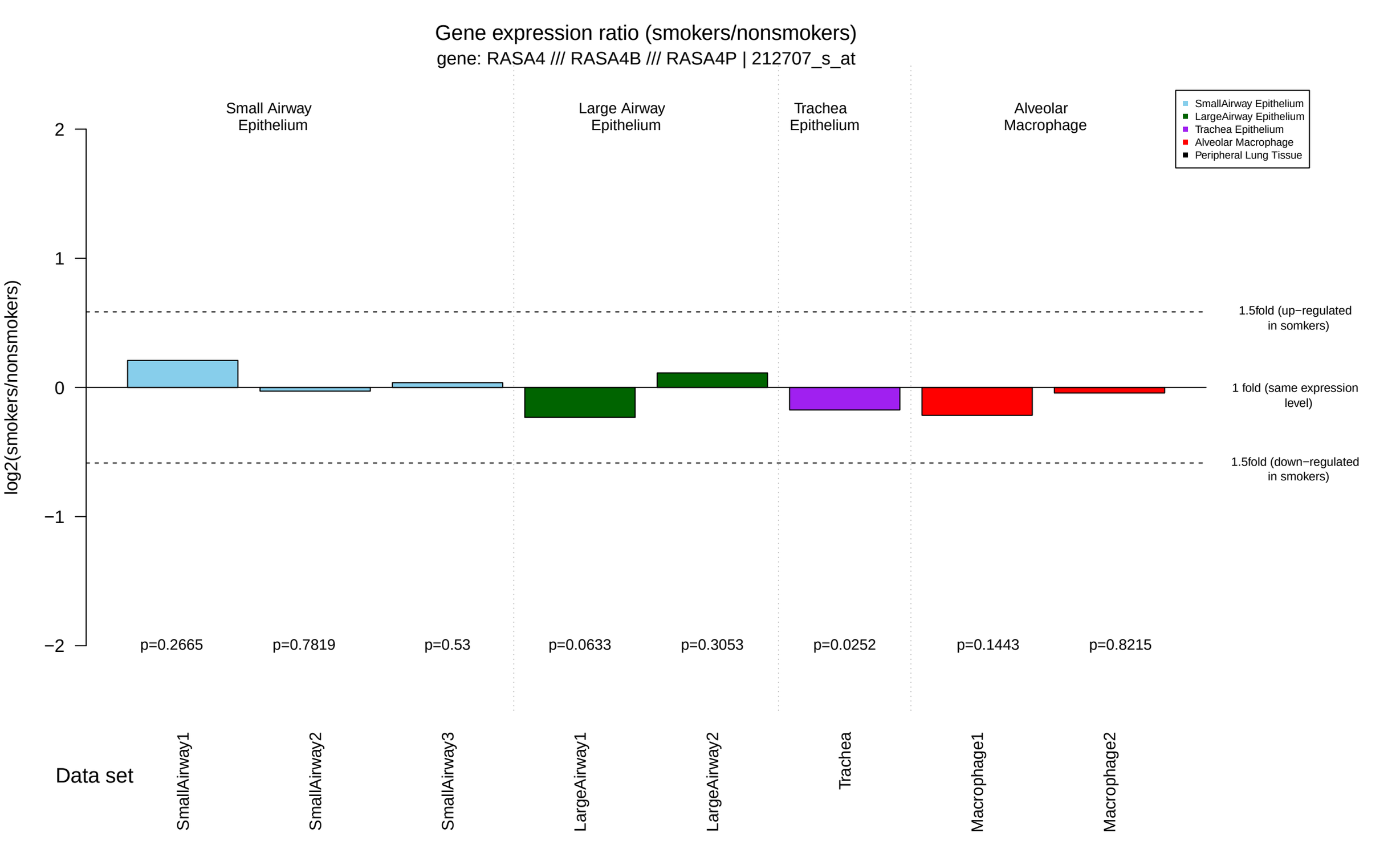
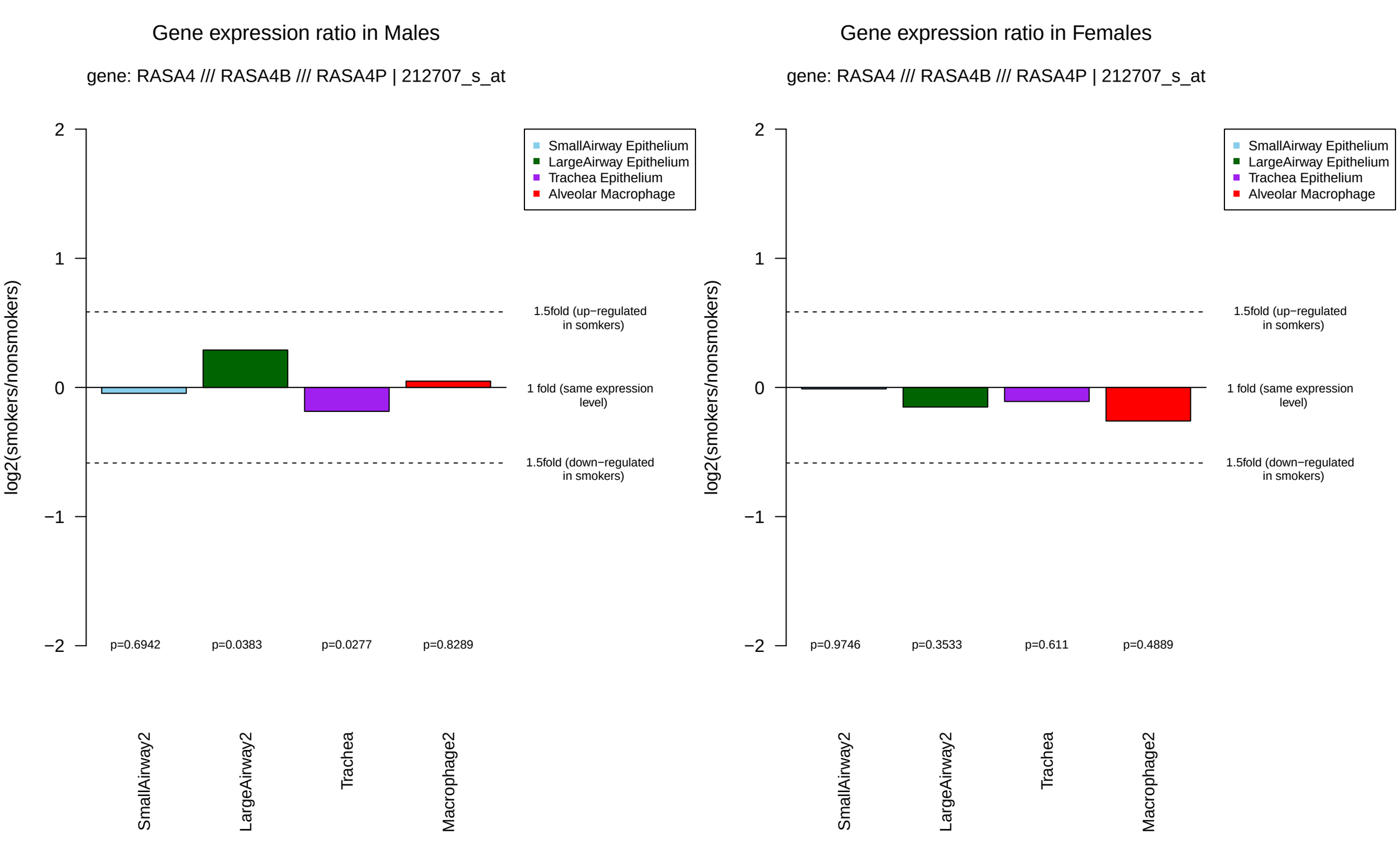
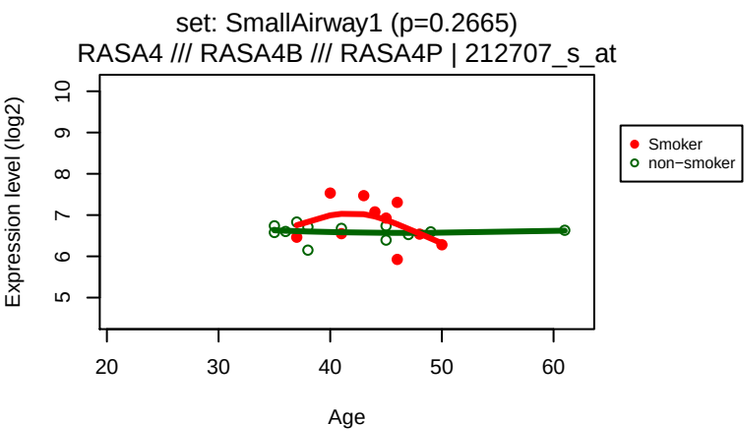
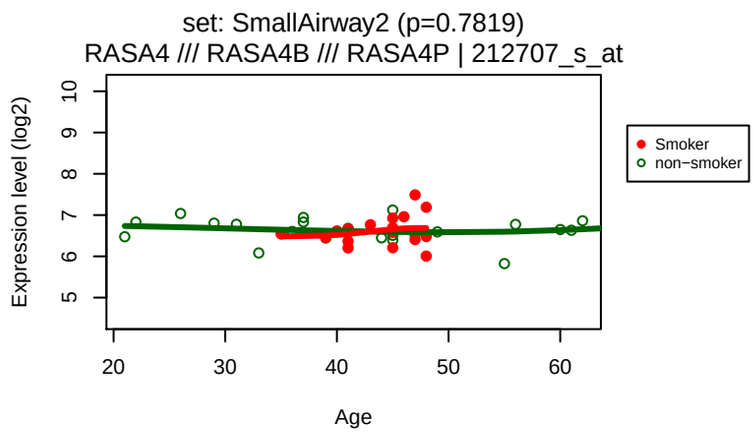
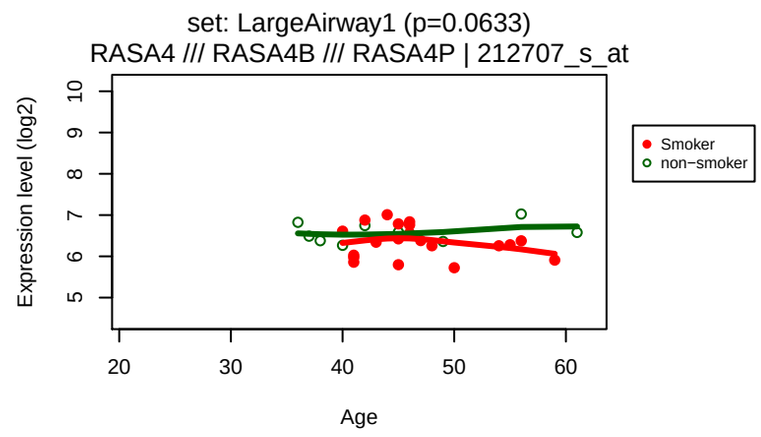

Gene expression of smokers and nonsmokers
gene: RASA4 /// RASA4B /// RASA4P | 212707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2665
p=0.7819
p=0.53
p=0.0633
p=0.3053
p=0.0252
p=0.1443
p=0.8215
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

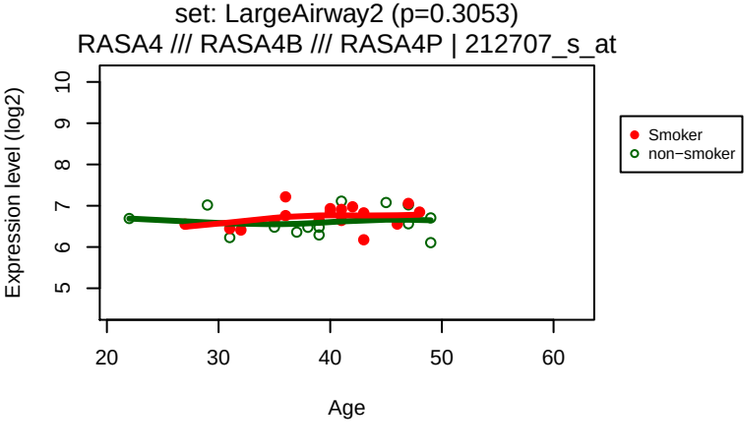

Gene expression ratio (smokers/nonsmokers)
gene: RASA4 /// RASA4B /// RASA4P | 212707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2665
p=0.7819
p=0.53
p=0.0633
p=0.3053
p=0.0252
p=0.1443
p=0.8215
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

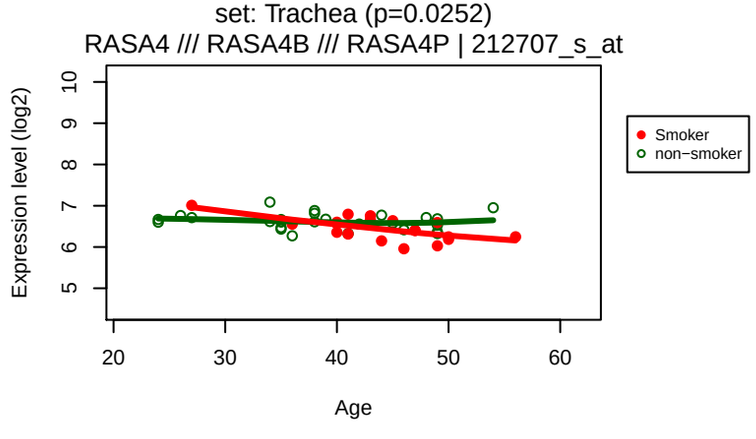

Gene expression ratio in Males
gene: RASA4 /// RASA4B /// RASA4P | 212707_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6942
p=0.0383
p=0.0277
p=0.8289
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASA4 /// RASA4B /// RASA4P | 212707_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9746
p=0.3533
p=0.611
p=0.4889
SmallAirway2
LargeAirway2
Trachea
Macrophage2

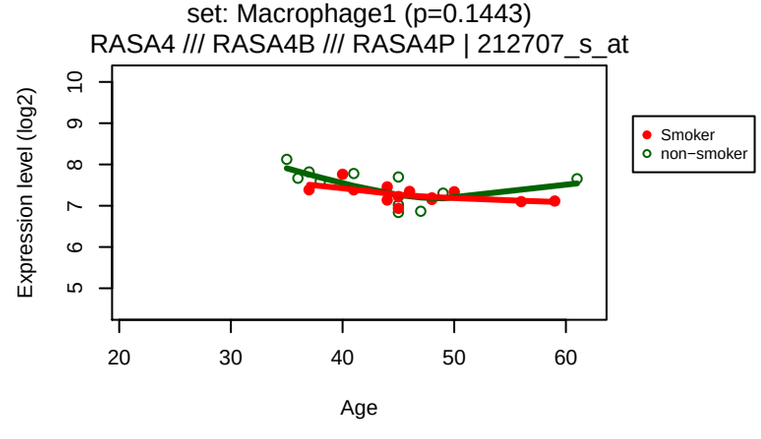

set: SmallAirway1 (p=0.2665)
RASA4 /// RASA4B /// RASA4P | 212707_s_at
Expression level (log2)
Age
Smoker
non−smoker

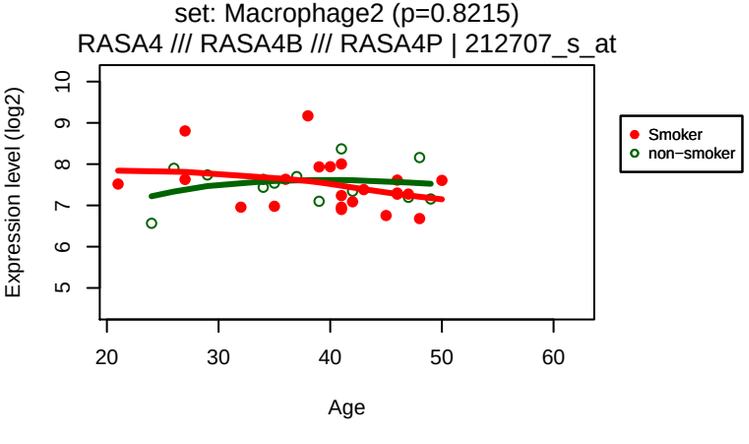

set: SmallAirway2 (p=0.7819)
RASA4 /// RASA4B /// RASA4P | 212707_s_at
Expression level (log2)
Age
Smoker
non−smoker

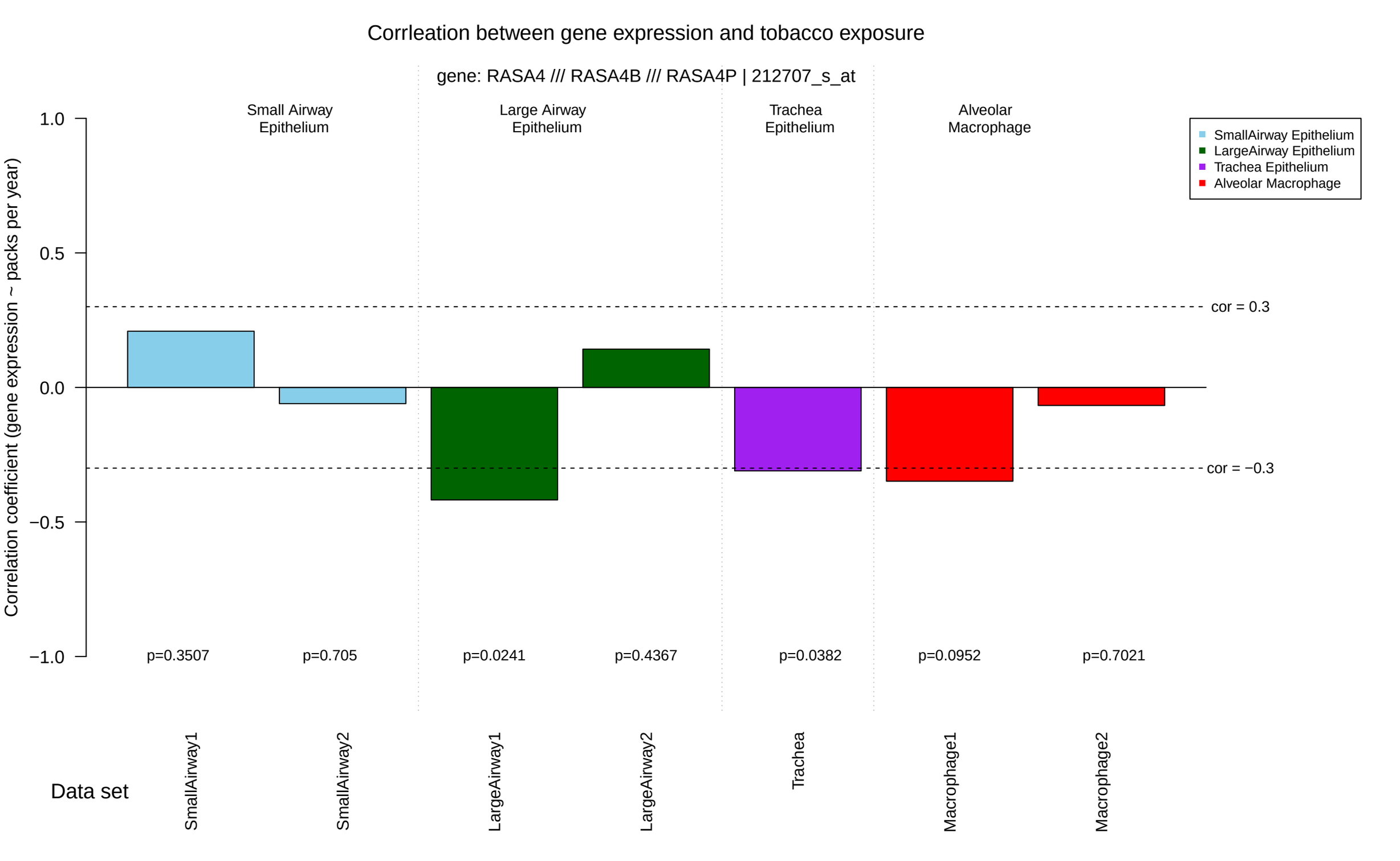

set: LargeAirway1 (p=0.0633)
RASA4 /// RASA4B /// RASA4P | 212707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3053)
RASA4 /// RASA4B /// RASA4P | 212707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0252)
RASA4 /// RASA4B /// RASA4P | 212707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1443)
RASA4 /// RASA4B /// RASA4P | 212707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8215)
RASA4 /// RASA4B /// RASA4P | 212707_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASA4 /// RASA4B /// RASA4P | 212707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3507
p=0.705
p=0.0241
p=0.4367
p=0.0382
p=0.0952
p=0.7021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2